



Air Quality Monitoring System



Introduction

OVERALL AIR POLLUTION:
85% UNHEALTHY

Problem Statement: Air pollution is a significant occupational hazard. Current industrial monitors are prohibitively expensive and lack real-time scalability.

System Architecture

Hardware: ESP32 MCU, MQ-135 Gas Sensor, SDS011 PM2.5 Sensor.
Power: 5V Supply.



A. Hardware Node Layout

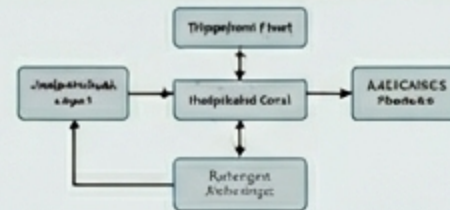


B. Network Architecture

Methodology & Data Processing

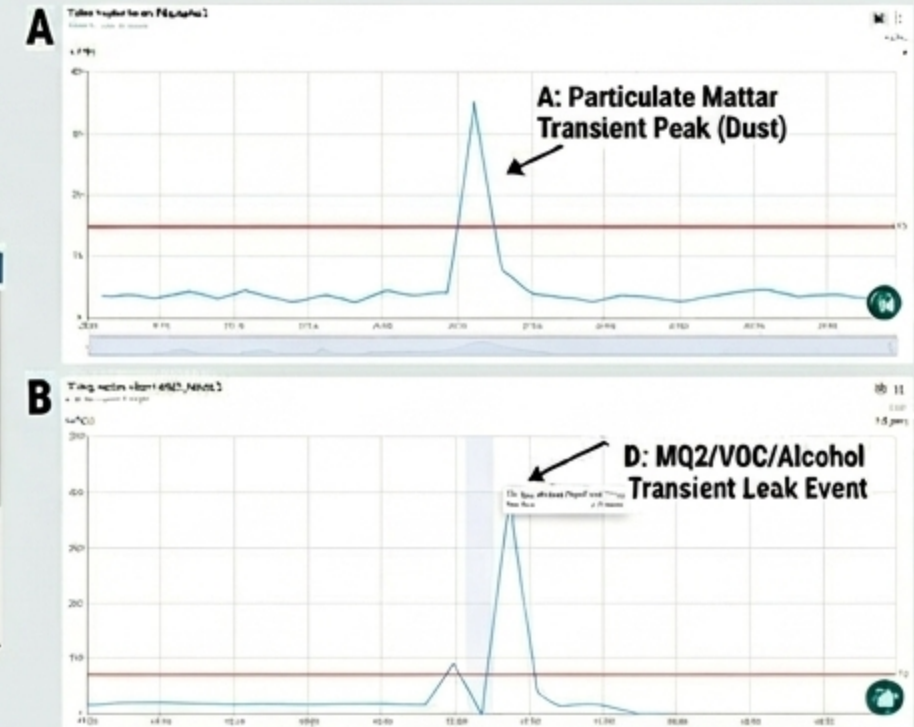
Embedded Firmware Flow:

1. Initialization.
2. Periodic Sensor Polling.
3. Data Calibration & Normalization.
4. JSON Formatted Data Transmission via Wi-Fi.



Price and performance (PPP) is often critical in embedded systems; however, our project has **prioritized and achieved specific project requirements** over generalized PPP metrics.

Results & Discussion



Demonstrated response to PM and VOC gas leak events. Data consistency to ThingsBoard Cloud.

Conclusions & Future Work

Future work:

- Integration of GPS for spatial heat-mapping.
- Enhanced multi-node network synchronization.
- Advanced sensor fusion algorithms.



<https://github.com/Eng-Mahdi47/Embedded-Systems>